

No technical knowledge required

The CheMatSustain Database

A guide for everyone: what it contains, how information gets there, who can see it, and how a scientific spreadsheet becomes a useful report.

For researchers, project partners, public viewers and data owners

Edition

1.0

Prepared

14 September 2026

Reading time

About 15 minutes

Companion document

Technical edition available separately

What is CheMatSustain?

CheMatSustain is a shared research information system for selected chemicals and nanomaterials. It helps project partners turn laboratory spreadsheets into organized, understandable reports.

In one sentence: the platform preserves the original research file, organizes its contents, controls what may be shared, and presents approved information through a website or controlled data connection.

Why does the project need it?



Different experiments

Laboratories carry out many kinds of tests, each with different measurements, units, tables and spreadsheet layouts.



One place to find results

The platform gives people a consistent way to find a material, understand a test and explore its results.



Controlled sharing

Data owners decide whether a test is public and which parts of it can be shown publicly.



Future reuse

Organized information is easier to compare, interpret and reuse in later research—with the right permission.

What kinds of tests are included?

The system supports 19 test types, including material characterization, biological response and environmental tests. Examples include DLS, FTIR, XPS, XRD, TGA, MTT, ROS, Trypan Blue, Water Flea and Algae tests.

This is more than a file library. The platform does keep the original spreadsheet, but it also turns the spreadsheet's contents into searchable information, tables, charts and reports.

How Information Moves Through the System

How information moves through CheMatSustain

The platform preserves the original research file, organizes the results, and shows only approved information.



THE IMPORTANT PRINCIPLE

Being listed publicly does not mean every result is public.

Test conditions, raw measurements, processed data, final results, and statistics can each be shared—or kept private—separately.

CheMatSustain · Plain-language guide

Figure 1. The main journey of a scientific record. Publication is a deliberate checkpoint, not an automatic result of uploading.

Five simple steps

1. **Upload:** an authorized administrator uploads a laboratory spreadsheet.
2. **Check and organize:** the platform reads the known spreadsheet layout and places information into consistent sections.
3. **Store:** it keeps both the original file and the organized record in protected storage.
4. **Choose what to share:** an administrator reviews publication settings for the test and each section.
5. **Use:** approved people see reports; approved computer systems can receive controlled data access.

Upload does not mean publish. A newly uploaded test is not automatically public. Publication is a separate choice, and each major part of the record has its own sharing switch.

What Does One Test Record Contain?

Although every scientific test is different, CheMatSustain organizes its information into five familiar groups.



1. Test details

The experimental context: material, preparation, method, equipment, cell line, treatment and similar conditions.



2. Raw data

The observations or measurements captured during the experiment, such as readings, runs or replicates.



3. Processed data

Values calculated or normalized from the original observations.



4. Final results

The summarized scientific outcome, often including averages and variation.



5. Statistical analysis

Formal analysis used to judge patterns or differences, such as analysis of variance and comparisons between conditions.

How do people find a record?

Three labels identify a test in the portal:

Work package

The project work area

Material identifier

The selected chemical or nanomaterial

Test name

The experimental method

Why keep both a record and a file? The organized record is best for finding and viewing results. The original spreadsheet preserves the source context and allows authorized teams to review or process it again later.

What Can the Public See?

Public sharing works at two levels: first, decide whether the test may appear publicly; second, decide which parts of that test may be shown.

Example: a public test with only selected results

The record can be listed publicly so people know that the test exists. The data owner might release the final outcome and statistics while keeping experimental conditions and detailed measurements private.



Test details



Raw data



Processed data



Final results



Statistics

A small identification header remains visible

For a public report to make sense, a limited set of identifying information can remain visible even when detailed test conditions are not released. This includes the test's name, short name, type, endpoint, reported endpoint outcome, procedure reference and reference-material identifier.

What does not happen: hidden information is not sent to the public browser and merely covered up on screen. The system removes unreleased sections before sending the public response.

Publication checklist for data owners

- ✓ Confirm that the correct material and test are selected.
- ✓ Review each section, not only the final result.
- ✓ Check whether raw measurements contain confidential context.
- ✓ Release only the sections approved under project policy.
- ✓ Open the public view and confirm it shows exactly what was intended.
- ✓ If the test must become private, unpublish it; its section-sharing choices are revoked.

Who Uses the Platform?



Public viewer

Creates a public account, signs in and sees only published tests and the sections approved for public release.

Does not see: private tests, hidden sections or internal file locations.



Team user

Uses an internally issued account and an email verification code. Sees internal research information they are authorized to access.



Administrator

Uploads and manages tests, controls publication, manages users and organizations, assigns access and reviews public access history.



Approved computer system

Uses its own issued credentials to request specific kinds of data. It receives only the permitted resources and capabilities.

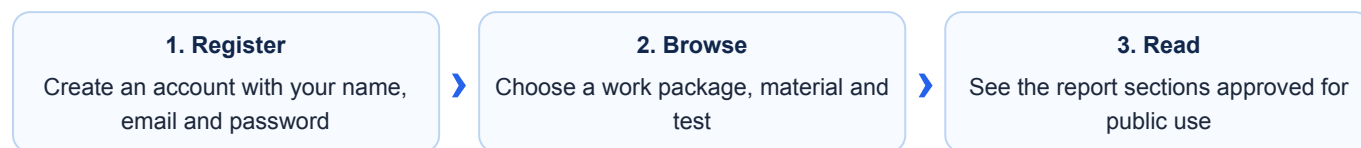
At a glance

Activity	Public viewer	Team user	Administrator
See published, released information	Yes	Yes	Yes
See authorized internal research data	No	Yes	Yes
Upload a scientific workbook	No	No	Yes
Choose public sections	No	No	Yes
Manage users and access	No	No	Yes

Accounts and automated access are separate. A person's website login is not used as a computer system's credential. This makes automated access easier to limit, rotate or disable without sharing a personal password.

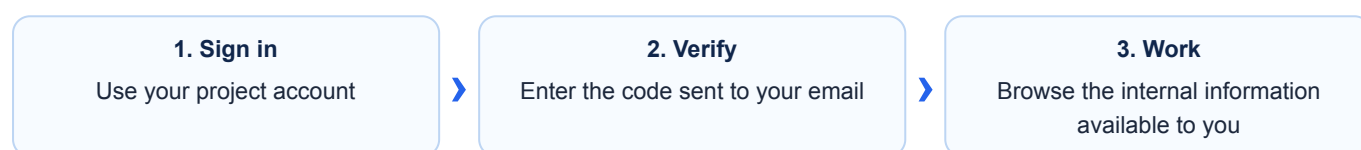
What Happens When You Use the Portal?

If you are a public viewer

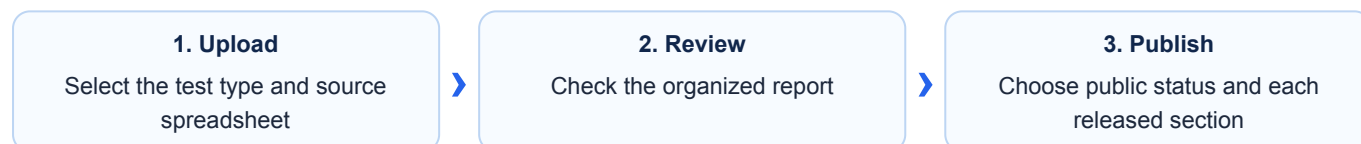


The platform records an access-history entry when an authenticated public viewer opens a test. This entry links the viewer, test, time and released sections. Live entries and their protected archives are kept for 20 days.

If you are a team user



If you are an administrator



If something looks wrong after upload: do not publish the record. Check that the correct spreadsheet template and test type were used, then ask a platform administrator or maintainer to review the parser warning or error.

Trypan Blue: From Spreadsheet to Report

Trypan Blue—shown as **TB** in the portal—is a useful example because its spreadsheet contains test conditions, repeated measurements, processed values, final results and statistics.

1 The spreadsheet

The laboratory workbook has recognizable sheets for test conditions, raw data runs, processed data runs, final results and, when available, statistics.

2 The platform reads known locations

The TB importer knows where expected labels and values are normally found. It also handles common spreadsheet differences such as dates, decimal commas and empty cells.

3 The information is organized

The platform groups material preparation, cell line and treatment details; collects each measurement run; associates processed runs; summarizes final conditions, repeats, averages and variation; and includes the statistical tables when present.

4 The source and record are saved

The original workbook is kept for traceability. The organized information becomes the searchable test record used by the portal.

5 The TB report is displayed

The viewer turns available information into clear sections, tables and charts. People can switch between runs and download table data where offered. If a section is not public, its tab does not appear to public viewers.

What might appear in a TB report?

- Material and dispersion information
- Cell-line and treatment conditions
- Percent cell death for chambers or conditions
- Repeated measurements and summarized averages
- Variation around the average
- Statistical comparisons between conditions

How Is Information Protected?

No single control does all the work. CheMatSustain uses several layers so that a mistake or failed control is less likely to expose data.



Secure sign-in

Passwords are not stored in readable form. Team users and administrators also confirm sign-in with an email code.



Roles and assignments

People and approved systems receive only the access appropriate to their role, organization and assigned resources.



Publication gates

Public status and each major information section are controlled separately. The server removes hidden content before responding.



Access history

Relevant public and automated data activity is recorded so access can be reviewed and attributed.



Protected files

Research workbooks and procedure documents are kept in protected storage, not in the public website files.



Checked releases

Automated security checks review secrets, code, dependencies, data separation and software images before deployment.

Privacy by response, not just by screen. The most important protection for public reports happens before information reaches the viewer: unreleased sections are left out of the response.

Good practice still matters. Users should not share accounts, passwords, email codes or issued computer credentials. Publication decisions should be reviewed against the project's data-governance policy.

Common Questions

Can anyone browse the public database without an account?

No. A public viewer registers and signs in. Their account remains limited to published, released information.

Does “public test” mean the complete spreadsheet is public?

No. The original spreadsheet remains protected, and the five information sections can be released separately.

Why can I see the report title but not the Test Conditions tab?

A small identification header remains visible so the report can be understood. Detailed conditions may still be withheld.

Why do team users receive an email code?

It adds a second sign-in step for accounts that can reach internal research information.

Can I download what I see?

Some scientific viewers offer downloads for displayed tables, charts or images. A download contains only information that was available in your authorized view.

What if a chart or table is empty?

The section may not have been present in the source workbook, may not apply to that test, or may not have been released to your role.

Can an uploaded test be corrected?

Yes. An administrator can replace the source file and update the organized record. The operation is designed to avoid leaving an unlinked old or failed file.

What should happen when somebody leaves the project?

Their website account should be disabled. Any separately issued automated-access credentials must also be disabled as part of offboarding.

Who Should I Contact?

Your situation	Best first contact	Useful information to provide
I cannot sign in or receive my email code.	Platform administrator	Your account email, approximate time and the screen message. Never send your password.
I need access to an internal test or protocol.	Your organization's data owner or platform administrator	Work package, material, test/protocol and business reason.
A test should be made public or private.	Authorized data owner / administrator	Exact test identity and the approval basis for each section.
An uploaded spreadsheet produced unexpected results.	Platform maintainer	Test type, workbook template/version, expected outcome and visible warning—without sending confidential data in an open channel.
My external software needs data access.	Platform administrator / API owner	Organization, intended use, required information and which tests/protocols are needed.
I suspect information was exposed or an account was misused.	Security contact immediately	Time, affected page/record, what you observed and any safe evidence. Do not investigate by accessing more data.

Remember these four points

1. Upload is not publication.

Sharing remains a separate, deliberate decision.

2. Public is not all-or-nothing.

Each major scientific section has its own release choice.

3. The source file stays protected.

Public reports use organized, approved information.

4. Access should be attributable.

Use your own account or assigned system credential.

Need technical detail? See the companion *CheMatSustain Database and Platform Documentation*, which contains the architecture, database catalogue, implementation relationships, operational controls and complete TB coding walkthrough.

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